

STAGE 5: CRITICAL COMPARATIVE REVIEW

CRITICAL COMPARISON

Honest self-evaluation of Stage 4 Report

The client's requirements and goals are at the centre of our proposed solution. The main issues identified were the inadequate functional capacity, accessibility and sustainability within the learning spaces. Our solutions, including modified HVAC systems, motion-sensor lighting, modular furniture, and accessibility enhancements, directly addressed these concerns. This alignment ensured that the client's requirements were effectively met.

The strongest aspects of our report are the detailed Risk Management Plan which comprehensively addresses all the risks from pre-installation to the maintenance. These risks are even colour-coded based on their severity, potential impacts and their respective mitigation plan are clearly outlined. The clearly-structured Stakeholder Engagement Plan outlining the purposes and communication strategies of all the relevant stakeholders at different stages of the project lifecycle along with robust technical depth and detailed implementation plan of the solutions developed.

We could have delved deeper into the financial feasibility analysis by accounting for the future costs in additional breakages or repairs thus, strengthening our report. We should have made the solution development section more concise and included visual representations and analyses to retain reader's attention.

Benchmarking against other report

The other team did a commendable job by coming up with distinct innovative solutions which comprise 3D-Printer Shelving, Vertical Storage Module to tackle storage constraints, redesigning the existing fixed student desks and introducing soundproof panels to enhance accessibility and students' learning experience (TUT07-GRP01, 2024). Upon further inspection, several differences became evident, including: the extensive use of diagrams and analysis to support the proposed designs provided a clearer visual representation in the solution development section of their report. This aligns with the principles outlined in the article "How Multimedia Can Improve Learning and Instruction" (Mayer, 2019), which underscores the benefits of multimedia in enhancing learning and understanding of content. Both the teams have adhered to all the requisite rubric assessment criteria in their reports. However, their report was exceptionally well-presented compared to ours. For example, their economic plan displayed a significantly higher level of detail, supported by relevant data. As a result, the peer's well-versed report was better in contrast.

Engineering Requirements and precedents

Our Engineering Requirements encompass both current and future designs that align with the client's needs for functionality, sustainability, and accessibility. While we focused on the upcoming solution, we did not include existing implementations, such as the special waste unit and updated lab equipment, in the report. By researching precedents from other institutions like Harvard, UTS Hive Lab, we have identified areas for improvement in our accessibility and sustainability measures, which are incorporated into our recommendations. Though our comprehensive proposed solution covers all the three aspects with realistic cost estimates well-under the given budget, I feel we could have come up with more innovative solutions to exceed client expectations.

In conclusion, the comparison of our work with the peer report has allowed me to assess the quality of the teamwork we displayed in Stage 4: Implementation Plan. While our report effectively addressed all the client's goals along with the risk management plan and stakeholder engagement plan, there are still areas that can be improved, particularly in the economic plan and solutions development sections.

REFLECTION

Team work & Time management

Reporting and responding

Our team (consisting of four members) worked really well throughout the period of the project, which was evident from the sparkplus peer review and tutor's feedback. After being chosen at random to join the team, we engaged in an interactive get-to-know-you session during the tutorial which really helped us to bond over things and understand one another. For each stage of the project, we divided our tasks equally in advance and I had the additional role of sending group reminders about deadlines.

Relating

During every stage of the project, I made it a priority to start my work early. When I was halfway done, I would remind others to start working on their parts. As a result, I often found myself making necessary adjustments to my section until the end in order to ensure our reports looked cohesive and maintained a seamless transition between different sections, avoiding unnecessary repetition of information which made reviewing my work feel rushed, especially when it came to including citations in this particular project. Despite this, I often voluntarily sent reminders in the group chat to remind them about their Sparkplus peer review tasks, as they tended to forget. I believe that these reminders helped us stay on track and complete all tasks before deadlines and meet the client requirements.

Reasoning

The quality of our work is reflected in the high-grades and positive feedback that we received from both staff and peers (Sparkplus, n.d.) (TUT05-GRP10, 2024) (TUT05-GRP10, 2024). While our autonomous allocation of tasks based on individual capabilities proved successful for team performance in this project, which can be observed in the article titled *"The Dynamical Relation Between Individual Needs and Group Performance: A Simulation of the Self-Organising Task Allocation Process"* (Wang et al., 2021). However, upon reflection, I believe there was room for improvement in our approach to time management. During my online search, I came across an article titled *"5 Crucial Benefits of a Well-Run Team Meeting"* (Bárcenas, 2019), highlighting the importance of regular team meetings in enhancing team performance. This resonated deeply with me, as we were unable to meet frequently outside of tutorial time which resulted in not having enough time for a thorough review of the entire report and ensuring that we addressed all client requirements.

Reconstructing

Starting with the next project, I will take full responsibility for organising regular team meetings, even if no one else steps up. By doing so, I will ensure that everyone is kept up to date with their progress which will significantly enhance our team's time management (Bárcenas, 2019). As I approach the group assignment in the upcoming unit SOFT2412: Agile Software Development Practices, I will go

the extra mile to create a well-structured schedule for weekly or fortnightly team meetings, tailored to accommodate all members. Additionally, I will consistently remind them about upcoming tasks and deadlines, ensuring that our team stays focused and has ample time to thoroughly work on the entire project before submission.

Getting additional information to strengthen our project

Reporting and responding

Before starting off with Stage 2 of the project, we had thorough discussions about potential solutions and their reliability. Later, we felt the need to visit these learning spaces - S170 Halliday's Lab and S270 Makerspace in-person to deeply understand the underlying problems being faced by our client. Though, we were able to have a friendly and insightful chat with one of the staff members in Makerspace and see the space in-person. However, we couldn't get any permission to visit the Halliday's Lab and therefore, we were only able to collect more information from feedback with two of the students which we then used as the base for the modifications in our solution.

Relating

Visiting the client site in person provided me and other team members with a comprehensive understanding of the current floor plan, equipment, and workstations being utilised in the S270 Makerspace area [Appendix 1][Appendix 2]. This allowed us to determine the feasibility of our initial ideas, discard the impractical ones and come up with more viable alternatives, as reflected in the Decision Matrix (reference) of Stage 2. Furthermore, engaging with individuals who participate in tutorials and frequent these learning spaces for their classes provided invaluable insights into the challenges they encounter, which I wouldn't have been aware of otherwise, given that I am not an AMME student.

Reasoning

By going that extra mile to gather additional information, we not only strengthened our project but also deepened our understanding of the challenges faced by both faculty and students. Additionally, the queries addressed by the client on Padlet (Goldfinch, 2024) (Goldfinch, 2024) and in the meetings provided a more comprehensive perspective which allowed us to meet our client's expectations. This aligns with the article I came across online titled "*An Empirical Investigation into Different Stakeholder Groups' Perception of Project Success*" (Davis, 2017) which suggests that collaboration between stakeholders and team is crucial for project success. Additionally, another article titled "*The Impact of Project Team Characteristics and Client Collaboration on Project Agility and Project Success: An Empirical Study*" (Radhakrishnan et al., 2022) emphasises a direct relationship between team characteristics, client collaboration, and project success measures such as on-time project completion, on-budget project completion, meeting project specifications, and project success rating.

Reconstructing

Reading the articles again made me realise the importance of effective communication between team and client to the success of a project. Consistent interactions foster transparency and keep all the involved parties updated on the project's progress and client's expectations (Radhakrishnan et al., 2022). In the next group assignment of the unit SOFT2412: Agile Software Development Practices, I am fully prepared to take it a step further by openly addressing any concerns with the team and staff in order to gain a deeper understanding of the project specifications. This will allow me to make more informed decisions and effectively address any potential issues that may arise.

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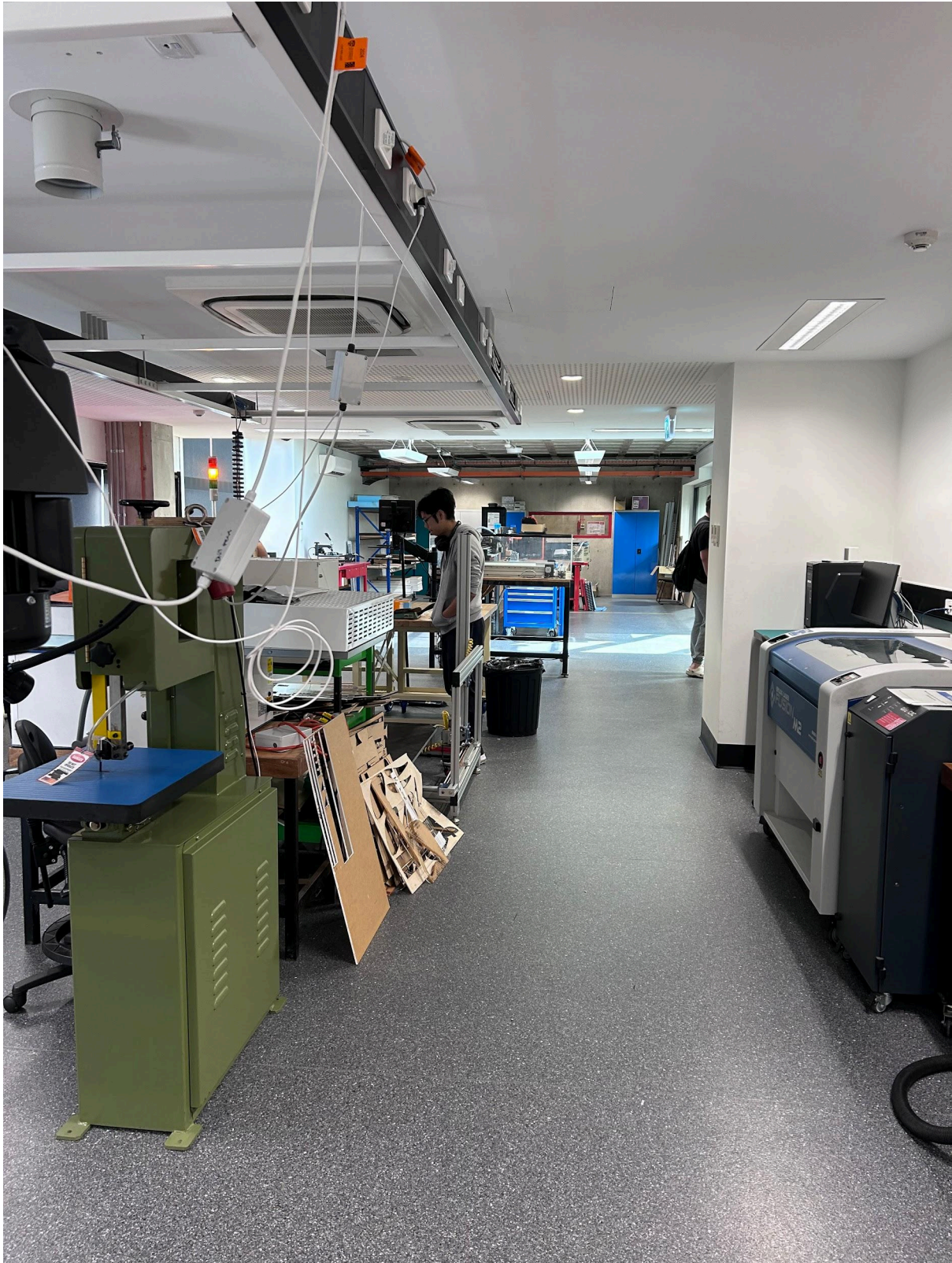
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APPENDIX

APPENDIX 1: IMAGE OF MAKERSPACE



APPENDIX 2: IMAGE OF MAKERSPACE

